

**AMENDMENT NO. 1 DECEMBER 2011
TO
IS 15103 : 2002 FIRE RESISTANT STEEL —
SPECIFICATION**

(Page 4, clause 10.3) — Substitute the following for the existing:

‘The tensile strength, yield strength and percentage elongation, when determined in accordance with IS 1608 or as per Annex A, shall be as given in Table 3.’

(Page 7) — Insert the following Annex A at the end of the text:

**ANNEX A
(Clause 10.3)**

**PROCEDURE FOR TESTING OF TENSILE SPECIMENS AT
ELEVATED TEMPERATURES**

It is well recognized that the strength of metallic materials depends a great deal on temperature. Therefore, while testing at elevated temperatures, close temperature control is necessary. For conducting tensile tests at elevated temperature for the determination of yield strength, tensile strength, elongation and reduction in area, the following testing procedure should be adopted. However, these procedures do not apply to the conditions of rapid heating or rapid strain rates.

A-1.0 Test Specimens and Sample

The size and shape of the test specimens should be based primarily on the requirements necessary to obtain representative samples of the material being investigated.

A-1.1 Unless otherwise specified, test specimens shall be oriented to the direction of fabrication.

A-1.2 At the centre, for products 38 mm or less in thickness, diameter or distance between flats.

A-1.3 Midway from the centre to the surface for products over 38 mm in thickness, diameter or distance between flats.

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A-1.4 The recommended dimensions of the standard as well as small size samples proportional to standard have been shown in Fig. 1 for flat samples and in Fig. 2 for round samples.

A-2.0 Temperature Control

A-2.1 The thermocouples must be kept in intimate contact with the specimen.

A-2.2 If the length of the reduced section of the specimen is less than 50 mm at least two thermocouples should be attached, one near each end. For reduced section more than or equal to 50 mm a third thermocouple should be attached near the centre of the specimen.

A-2.3 For the duration of the test, the difference between the indicated and nominal test temperatures should not exceed $\pm 3^{\circ}\text{C}$ up to and including $1\,000^{\circ}\text{C}$ and $\pm 6^{\circ}\text{C}$ above $1\,000^{\circ}\text{C}$. Internal heating due to plastic deformation may add to the rise of temperature. In such cases, temperature at maximum force must be recorded and reported.

A-2.4 Temperature overshoots during heating should not exceed the limits specified in **A-1.3**, unless agreed upon by the customer and the supplier. The heating characteristics of the furnace and the temperature control devices should be studied to determine the power input, temperature set point, proportioning control adjustment and the placement of the control thermocouple necessary to limit transient temperature overshoots. It is desirable to stabilize the furnace at a temperature from 6 to 28°C below the nominal test temperature before making the final adjustments. If allowed, overshoots shall be reported with details of magnitude and duration.

A-2.5 The specimen must be held for sufficient duration prior to the start of the test in order to ensure that the specimen reaches equilibrium temperature and that the temperatures can be maintained within the limits specified in **A-1.3**. Unless otherwise specified, this time should not be less than 20 minutes. The time to attain test temperature and the time for which the specimen was held before testing should be reported.

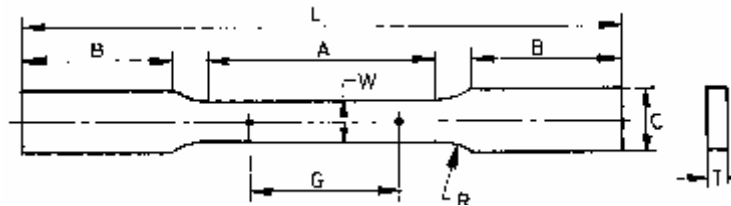
A-2.6 Care should be taken not to introduce non-axial forces while mounting the specimen. If packing is used to seal the furnace, it must not be so tight that the extensometer arms or pull rods are displaced or their movement restricted.

A-3.0 Strain Measurement and Strain Rate

Tensile properties of materials tested at elevated temperatures are, in general, affected by the rate at which strains are imposed during testing. Therefore, it is important to control and report the strain rate.

A-3.1 The strain rate in the uniform section of the test specimen during determination of yield strength should be maintained at $0.005 \pm 0.002 \text{ min}^{-1}$. After the determination yield strength, the cross head speed should be 0.05 ± 0.01 times the adjusted length of the reduced section of the specimen per minute. In cases where it is desired to establish the entire engineering stress strain curve of a specimen, the speed of testing may be maintained at 0.005 min^{-1} throughout the entire test. Non-contact methods of extensimetry have been shown to work well for this type of testing.

A-3.2 When yield strength determination is not required, an extensometer need not be used. A pacing device should be used to maintain the rate of crosshead motion at 0.05 ± 0.01 times the adjusted length of the reduced section of the specimen per minute.



Dimensions			
Nominal Width	Standard Specimens		Subsize Specimen
	Plate-Type 40 mm	Sheet-Type 12.5 mm	6 mm
G — Gauge length	200.0 ± 0.2	50.0 ± 0.1	25.0 ± 0.1
W — Width	40.0 ± 2.0	12.5 ± 0.2	6.0 ± 0.1
T — Thickness	Thickness of material		
R — Radius of fillet, <i>Min</i>	25	12.5	6
L — Overall length	450	200	100
A — Length of reduced section, <i>Min</i>	225	57	32
B — Length of grip section	75	50	30
C — Width of grip Section	50	20	10

FIG. 1 DIMENSIONS RECOMMENDED FOR STANDARD FLAT SAMPLE. DIMENSIONS FOR SMALL SIZE SAMPLES PROPORTIONAL TO STANDARD SAMPLE HAVE ALSO BEEN SHOWN

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Dimensions, mm					
	Standard Specimen		Small-Size Specimens Proportional to Standard		
	12.5	9	6	4	2.5
<i>G</i> — Gauge length	62.5 ± 0.1	45.0 ± 0.1	30.0 ± 0.1	20.0 ± 0.1	12.5 ± 0.1
<i>D</i> — Diameter	12.5 ± 0.2	9.0 ± 0.1	6.0 ± 0.1	4.0 ± 0.1	2.5 ± 0.1
<i>R</i> — Radius of fillet, <i>Min</i>	10	8	6	4	2
<i>A</i> — Length of reduced section, <i>Min</i>	75	54	36	24	20

FIG. 2 DIMENSIONS RECOMMENDED FOR STANDARD ROUND SAMPLE. DIMENSIONS FOR SMALL SIZE SAMPLES PROPORTIONAL TO STANDARD SAMPLE HAVE ALSO BEEN SHOWN

(MTD 4)

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